

The Bridge to Post-Quantum: Benchmarking Hybrid Cryptographic Implementation

M. Danish Raza (22K - 4183)

Talha Muneer Saeed (22K - 4465)

Section: 8 - A

Submitted To: Dr. Sumaiyah Zahid

Abstract – Problem & Response

Problem

Shor's algorithm breaks integer factorization (RSA/ECC) in polynomial time, threatening classical security.

Response

PQC standards (FIPS 203 / ML-KEM) and Hybrid KEMs are recommended to bridge to quantum-safe systems.

This Study

Part 1: Qiskit gate-level Shor simulations and resource extrapolation. Part 2: 1000-run empirical benchmarks of RSA-KEM, ML-KEM-512, and Hybrid KEM.

Introduction & Literature Gap

Quantum-safe migration is urgent — but the evidence base is still incomplete.

- **What existing work does well:** Strong guidance already supports moving to CRYSTALS-Kyber (ML-KEM) and hybrid key exchange.
- **What's missing:** Few studies show open, gate-level Shor threat models or reproducible benchmark data on real hardware.
- **Why it matters:** Without both threat estimates and performance costs, organizations can't plan a practical transition.
- **What this research adds:** A side-by-side view of quantum attack feasibility and PQC migration overheads.

Approach: We combine a Qiskit Shor circuit analysis with liboqs compilation and benchmarking on consumer hardware.

Methodology – Part 1: Shor Circuit & Gate-Level Modeling

01

QFT / IQFT Design

Gate primitives: H + controlled-phase rotations ($\theta = 2\pi/2^k$). Swaps reverse byte-order; IQFT uses negative rotations.

02

Modular Unitaries

Modular-exponential matrix via controlled permutation unitaries $U_{a^{2^i}}$; register sizing derived from $m = \lceil \log_2 N \rceil$ and $t = 2m$.

03

Assembly & Post-Processing

We measure the counting register after the circuit runs.

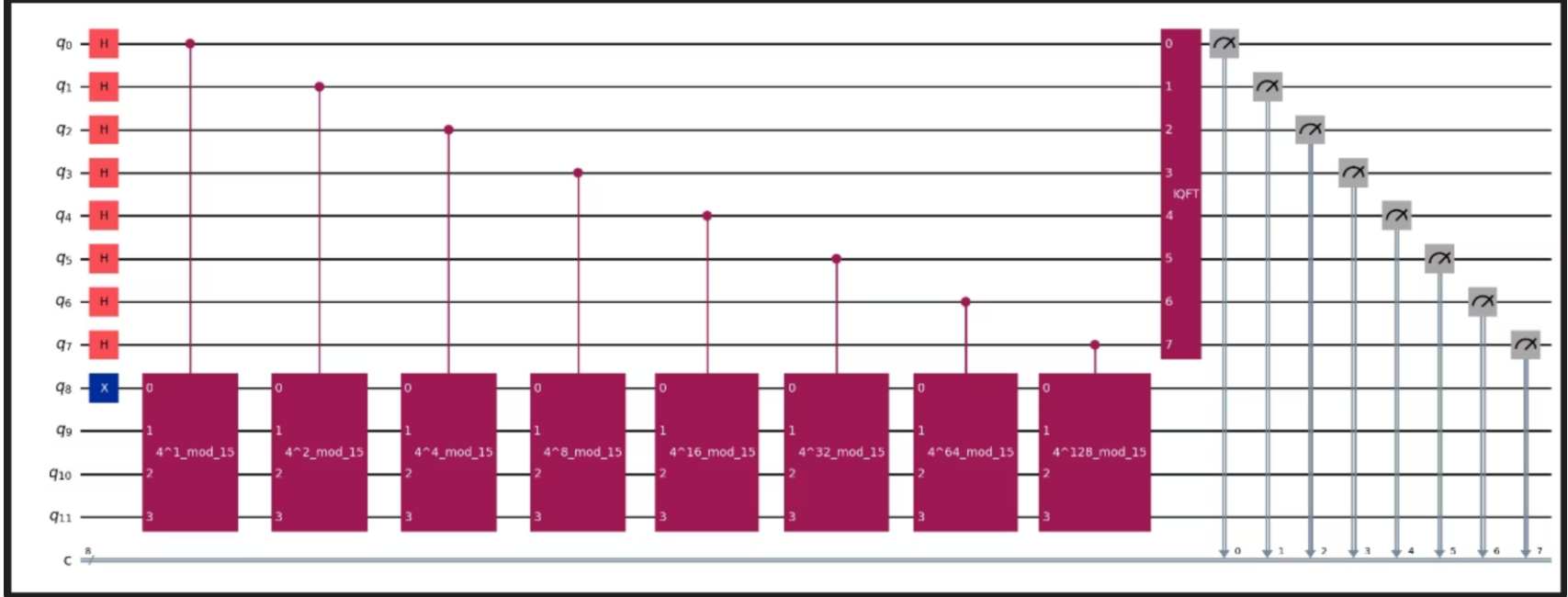
We use the result to estimate the period.

Then we recover factors with a gcd calculation.

```
g1 = math.gcd(pow(a_used, half, N) - 1, N)
```

```
g2 = math.gcd(pow(a_used, half, N) + 1, N)  
1, N)
```

$N=15$, $a=4$, $t=8$ counting qubits, $m=4$ target qubits, total=12



Shor's Circuit: for $N = 15$.

Methodology – Part 2: PQC Benchmarking & Hybrid Construction

Testing Methodology

Each operation was benchmarked over 1000 runs with fixed inputs and repeated configurations.

Timing used high-resolution wall-clock measurements at microsecond precision.

Results report mean $\pm$ standard deviation with warmup, cooldown, and isolated test conditions.

RSA-KEM Wrapping

RSA-2048 used PKCS#1 OAEP.

Each run wrapped a 32-byte random secret.

We measured both encapsulation and decapsulation.

Hybrid KEM

RSA and ML-KEM ran in parallel.

Public keys and ciphertexts were concatenated.

The shared secret came from $\text{SHA-256}(\text{ss_RSA} \parallel \text{ss_ML-KEM})$.

Shor Results – Spatial & Temporal Projections

Simulations for $N=15,21,35$ ($n=4,5,6$) produced a model projecting qubits and gates up to RSA-2048.

Qubit Projection

Empirical projection: 6,144 logical qubits. Optimized theoretical model: 4,099 logical qubits – aligns with prior estimates ($\sim 4k$).

Gate Count & Runtime

Gates $\approx 2 \cdot n^3 \Rightarrow$ RSA-2048 $\approx 1.72e+10$ logical ops.
Runtime: superconducting ($\tau=1 \mu s$) ≈ 4.77 hours;
trapped-ion ($\tau=100 \mu s$) ≈ 19.88 days.

PQC Benchmarks – 1000 Runs Summary

Key Generation

RSA-2048: 1150 ± 12.30 ms.

ML-KEM-512: 0.0151 ± 0.0034 ms

($\approx 5700\times$ faster).

Encapsulation / Decapsulation

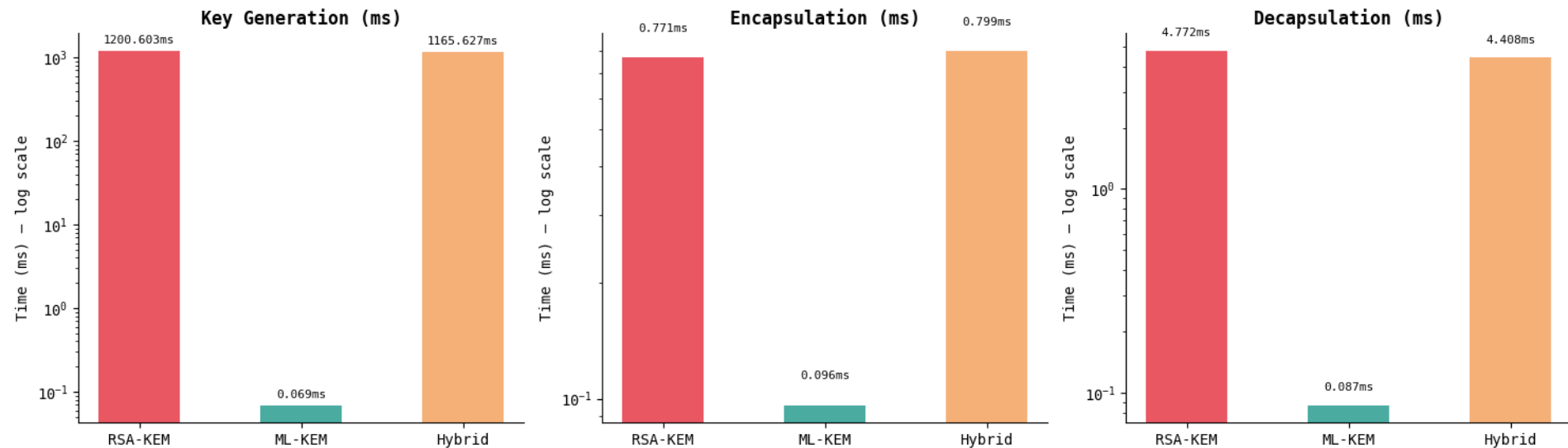
Encap – RSA: 0.1705 ± 0.0133 ms; ML-KEM:

0.0716 ± 0.0076 ms ($\approx 2.4\times$ faster). Decap –

RSA: 1.9961 ± 0.0811 ms; ML-KEM: $0.0673 \pm$

0.0069 ms ($\approx 30\times$ faster).

(mean $\pm$ std dev, 1000 runs, consumer hardware)



Transmission Sizes & Hybrid Overhead



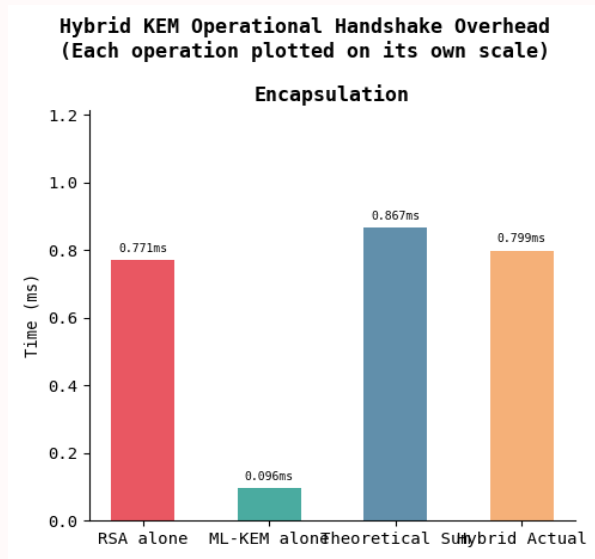
Public Key Sizes

RSA-2048: 294 bytes. ML-KEM-512: 800 bytes (0.78 KB).



Ciphertext Sizes

RSA-2048: 256 bytes. ML-KEM-512: 768 bytes.



Hybrid Viability

Hybrid encapsulation latency: 0.2443 ± 0.0189 ms — a negligible millisecond addition, confirming practical deployability.

Conclusions & Future Work

Threat is Quantifiable

Shor can feasibly break RSA-2048 with ~4k logical qubits and under 5 hours on superconducting hardware.

Hybrid as a Bridge

KDF-based Hybrid KEM provides negligible overhead and dual-layer security for transition planning.

PQC is Practical

ML-KEM-512 outperforms RSA in key ops and maintains compact transmission sizes — ready for production adoption.

Next Steps

Model physical error-correction overhead (surface codes → physical qubits) and test PQC under constrained network topologies.

References

1. Kinyua, S. (2025). "Transitioning to Post-Quantum Cryptography..." European Journal of Theoretical and Applied Computer Sciences, Vol. 3(1), pp. 45–58.
2. Gidney, C., & Ekerå, M. (2021). "How to factor 2048-bit RSA integers in 8 hours using 20 million noisy qubits." Quantum, 5, 433.
3. NIST (2024). "FIPS 203: Module-Lattice-Based KEM Standard."
4. Open Quantum Safe (OQS) Project. liboqs: <https://github.com/open-quantum-safe/liboqs>